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PROFESSIONAL SUMMARY

Frontend Engineer with 2 years of production experience building enterprise-grade React and TypeScript applications. Specialised in real-time data visualisation, shared component libraries, and performance-critical interfaces. Proven track record of improving reliability, cutting regression risk, and shipping maintainable code in Agile teams. Actively expanding into backend and systems engineering.

PROFESSIONAL EXPERIENCE

Formulatrix — Frontend Software Engineer
Jul 2024 – May 2026 · Surakarta, Indonesia

- Increased feature delivery reliability for laboratory automation workflows, reducing production bug rate by ~30%, by enforcing TypeScript strict mode and a modular, domain-scoped component architecture across a multi-team React codebase.
- Sustained 99%+ uptime for a production real-time visualisation platform serving robotic fleet monitoring, by maintaining and debugging React, Three.js, and SignalR integrations under high-frequency WebSocket data loads.
- Improved application reliability and reduced user-facing errors, cutting incident resolution time significantly, by diagnosing and fixing race conditions and state desync bugs in event-driven SignalR pipelines.
- Reduced UI re-render overhead and improved perceived responsiveness, by refactoring Redux state slices, extracting memoised selectors, and applying React.memo boundaries across visualisation-heavy components.
- Improved release confidence and reduced regression incidents, increasing unit test coverage by ~35%, by building and maintaining Jest and React Testing Library suites across shared component libraries.
- Accelerated onboarding for new frontend engineers, by documenting component APIs and contributing architectural decision records within an Agile/Scrum team of 10+.

KEY PROJECTS

Rover – Real-Time Robotic Fleet Visualisation Platform
[React] [TypeScript] [Three.js] [SignalR] [WebSocket] [Vite] [Redux] [Jest] [C#] [ASP.NET Core] [SQLite]

- Enabled continuous lab-operator visibility into a fleet of laboratory robots, sustaining real-time spatial rendering at 60 fps, by optimising Three.js scene graph updates and throttling SignalR event bursts on the frontend.
- Reduced production incident impact on end-users, cutting mean time to recovery by delivering targeted hotfixes under live-system constraints in a zero-downtime deployment pipeline.
- Reduced lint errors and positive-case build failures by ~30% per release phase, by implementing a CI/CD pipeline in Jenkins.
- Improved long-term maintainability of a complex visualisation codebase, by decomposing monolithic scene components into focused, testable units and introducing typed event contracts between the frontend and backend SignalR hub.
- Strengthened data integrity between UI and hardware state, by implementing optimistic UI patterns with rollback and reconciling WebSocket payloads against a normalised Redux store.

Fmlx Common UI – Internal Shared Component Library
[React] [TypeScript] [MUI] [Jest] [React Testing Library] [Storybook] [Vite]

- Accelerated UI development velocity across 3+ internal products, reducing duplicated component code by ~40%, by designing a shared React component library with consistent prop APIs, theming tokens, and full TypeScript generics support.
- Improved cross-team component adoption and reduced integration bugs, by co-authoring library documentation and usage guidelines consumed by 8 company product frontend applications.
- Kept consumer bundle sizes down, by maintaining tree-shakeable module exports and build-time optimisation across the library.
- Raised component reliability and prevented regressions across product releases, reaching 86% coverage including regression tests for legacy components still used by other company projects, by writing comprehensive Jest and React Testing Library suites covering rendering, interaction, and accessibility edge cases.
- Improved design-to-code consistency, by collaborating directly with UX designers to translate Figma design tokens into a centralised MUI theme extension used organisation-wide.

ESS – Internal Employee Self-Service Portal
[React] [TypeScript] [REST API] [JWT] [RBAC] [PostgreSQL] [C#] [ASP.NET Core]

- Delivered secure, role-based access control and multi-step authentication flows for an employee portal serving the full organisation, by implementing JWT-based auth with refresh-token rotation and a React Context permission layer.
- Reduced time-to-feature for HR and admin workflows, by building reusable data-table, form-validation, and notification components consumed across 6+ portal modules.
- Improved maintainability and unblocked new feature requirements, by migrating the portal's React components from a class-based to a function-based paradigm.

- Delivered a messaging feature for the Review application, by collaborating with the backend development team on API contract and database schema design.
- Improved backend integration reliability, by defining typed REST API contracts shared between the ASP.NET Core backend and React frontend, catching schema mismatches at compile time.

Promtomat – Binary Studio BSA 2026 Bootcamp Project
[React] [TypeScript] [Vite] [Ant Design] [RTK Query] [pnpm]

- Strengthened account security for a multi-user business platform, by implementing a password reset flow with secure token TTL and session invalidation.
- Reduced ambiguous failure states for end users, by building centralised HTTP error handling that distinguished 401 (unauthenticated) from 403 (unauthorized) responses, including a dedicated no-access UI.
- Kept a multi-contributor monorepo shippable under CI, by enforcing conventional commits via commitlint and resolving merge conflicts across concurrent feature branches in a pnpm workspace.

EDUCATION & TRAINING

RevoU — Fullstack Software Engineering Program
Oct 2023 – Apr 2024

- Shipped a full-stack project end-to-end using React, Python, Flask, SQLAlchemy ORM, and PostgreSQL within a team-based, sprint-driven workflow.
- Implemented a CI/CD pipeline on the project repository, automating build and linting and inspecting code quality.
- Built an SMTP server for email-based authentication code verification.
- Implemented authentication systems, form validation, REST API integration, and responsive UI across the project.

SMK St. Mikael Surakarta – Mechanical Engineering
2016 – 2019

TECHNICAL SKILLS

Frontend: React, TypeScript, JavaScript, Next.js, Redux, RTK Query, MUI, Ant Design, Tailwind CSS, Three.js, WebSocket, SignalR, Vite

Testing: Jest, React Testing Library

Backend: Node.js, Express.js, Python, Flask

Databases: PostgreSQL, MySQL

Tools & Infra: Git, Docker, ESLint, CI/CD pipelines, pnpm, commitlint

Methodologies: Agile, Scrum, TDD, Component-Driven Development